

Optimisation of agrivoltaic systems within the water-energy-food nexus

Abstract

Agrivoltaic (APV) systems enable the co-production of food and solar energy, yet their optimal design depends on complex interactions among climate, crops, and policy frameworks. This study introduces a multi-objective optimisation framework that integrates multi-year crop rotation, diverse APV configurations (vertical, one-axis tracking, overhead), and explicit national regulatory constraints across three European countries: Sweden, Germany, and Italy. The model couples photovoltaic (PV) and crop-growth simulations to evaluate trade-offs across the water–energy–food (WEF) nexus. Results show that both climate and regulation strongly shape feasible system geometries. Under Swedish subsidy limits (≤ 10 % land occupation), one-axis tracking systems achieved total land equivalent ratios (LERs) around 1.25 and water savings near 10 %. Under German yield-retention rules (≥ 66 %), overhead systems achieved LER around 1.36 and strong economic returns. While under Italian ministerial constraints combined with best practices, feasible designs emerged only in the enhanced WEF scenario (accounting for crop adaptation), achieving LER up to 1.66 with water-use reductions approaching 20 %. These outcomes demonstrate that applying identical policy thresholds across different climates can suppress overall sustainability performance, highlighting the need for climate- and crop-sensitive APV policy frameworks to unlock their full WEF potential.

Graphical abstract



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1. Introduction

The water–energy–food (WEF) nexus has emerged as a central framework for tackling global sustainability challenges, emphasising the interdependence of natural resource systems and the importance of integrated management strategies ([Gao et al., 2024](#); [FAO, 2024](#); [Campana et al., 2018](#)). Within this context, agrivoltaic (APV) systems, where photovoltaic (PV) panels are co-located with agriculture, have gained increasing attention as a means to advance all three pillars of the WEF nexus ([Barron-Gafford et al., 2019](#)). By enabling simultaneous solar energy conversion and crop cultivation on the same land, APV systems offer a dual-use model that enhances land-use efficiency while addressing multiple sustainability objectives ([Jain et al., 2021](#); [Zhang et al., 2023](#); [Cheng et al., 2022](#)).

Beyond simple land sharing, APV systems can generate synergistic benefits that amplify their value in climate- and water-stressed environments. The partial shading from PV panels can reduce soil evaporation and moderate local microclimates, thereby improving water retention and shielding crops from heat extremes ([Mamun et al., 2022](#); [Cuppari et al., 2021](#)). These effects are particularly advantageous in arid and semi-arid regions, where conserving water and maintaining food production are critical ([Ayadi et al., 2024](#)). In addition, APV systems contribute to climate mitigation by generating renewable electricity, thereby reducing dependence on fossil fuels ([Elamri et al., 2018](#)).

Several case studies illustrate the potential of APV systems across diverse geographic contexts. In Shenzhen, China, rooftop-based APV projects have demonstrated the feasibility of producing food and electricity in urban settings, potentially meeting local vegetable demand while providing clean energy ([Jing et al., 2022](#)). European field trials have reported land equivalent ratios (LERs) well above 1.0, sometimes approaching 1.8, indicating that combined outputs of food and electricity can exceed those achieved

through separate land uses ([Trommsdorff et al., 2021a](#)). However, such gains are highly site-dependent, shaped by climate, crop type, system configuration, and regulatory environment.

Despite these advantages, APV implementation faces key challenges related to system design and policy constraints. A central design dilemma arises from the trade-off between shading and light availability: while shading can protect crops and conserve water, it also limits photosynthetically active radiation (PAR), potentially reducing yields ([Jain et al., 2021](#)). The severity of this trade-off depends on the specific crop, panel configuration (height, row spacing, and orientation), and prevailing environmental conditions ([Dinesh and Pearce, 2016](#); [Sarr et al., 2024](#)). Shade-tolerant crops, such as leafy vegetables ([Marrou et al., 2013](#); [Laub et al., 2022](#)), may benefit from partial shading, whereas shade-sensitive crops can experience yield penalties if light availability is compromised ([Dupraz et al., 2011](#)). These sensitivities highlight the importance of crop selection and climate-aware design.

Recent research has shown that system geometry plays a critical role in balancing these competing demands. For example, vertical or east–west oriented PV arrays may improve light distribution across the crop surface but often at the expense of total energy conversion ([Kwon et al., 2020](#)). Conversely, south-facing arrays maximise solar output but can intensify shading on crops ([Asa'a et al., 2025](#)). These findings underscore the need for multi-objective optimisation in APV system design, where trade-offs between energy yield, agricultural output, and water use must be jointly considered ([Willockx et al., 2024](#)).

In addition to physical and biological constraints, policy and regulatory frameworks play a pivotal role in shaping APV feasibility. Many jurisdictions impose restrictions on the proportion of land that can be occupied by PV panels, the maximum height of structures, or the minimum electricity output required for grid integration ([DIN SPEC 91434, 2025a](#); [MASE, 2023](#); [Jordbruksverket, 2024](#); [Vidotto et al., 2024](#); [Dupraz, 2024](#)). These

policies are often introduced to balance agricultural preservation with renewable energy goals but can inadvertently constrain the design space and limit overall sustainability outcomes. Moreover, regulations vary widely between countries, and few studies have assessed how applying different policy constraints across diverse environmental conditions affects APV system performance.

Another limitation in current APV research is the frequent focus on a single location, crop, or year, which limits generalisability. Performance of APV systems is not only spatially variable but also temporally dynamic, interannual variability in weather conditions ([Sturchio and Knapp, 2025](#)), crop rotation patterns, and seasonal growth cycles can significantly affect outcomes. For instance, a system optimised under a typical meteorological weather year may underperform during drought years, or configurations suited for one crop may not support others in a multi-year rotation. Addressing these temporal and spatial dynamics requires more holistic models that reflect the complexities of real-world agricultural systems ([Trommsdorff et al., 2021b](#)).

The analysis focuses on Sweden, Germany, and Italy, which were selected to represent contrasting climatic and regulatory contexts and to extend previous work on APV shading effects and techno-economic performance ([Zainali et al., 2023](#); [Zidane et al., 2025](#)). To our knowledge, this is the first optimisation framework that combines multi-year simulations, multiple system configurations, and crop rotation within a unified platform under explicit national regulatory constraints ([Bellone et al., 2024](#); [Campana et al., 2021](#); [Miljkovic et al., 2024](#)). Previous studies have focused on individual configurations, including vertical APV in Sweden ([Campana et al., 2021](#)), one-axis tracking in the United States ([Miljkovic et al., 2024](#)), and several layouts in Italy ([Bellone et al., 2024](#)). The Swedish and US studies performed optimisation but did not include policy or regulatory constraints. The Italian study included a limited evaluation of row pitch using three fixed spacing steps for each system but did not systematically assess the effects of row pitch, azimuth, or height. None of these studies incorporated crop rotation. To advance APV system design toward more context-sensitive and policy-

aligned solutions, a comprehensive multi-objective optimisation framework was developed, integrating crop growth modelling, PV energy performance simulation, and regulatory constraints. The model captures interannual variation through a six-year crop rotation and explores a broad design space defined by system height, row spacing, and panel orientation. A composite WEF index is introduced to assess performance across sustainability domains, providing a structured basis for comparing trade-offs.

2. Methods

2.1. Study site and climate data

This study evaluated APV performance across three European locations: Kärrobo Prästgård (Sweden), Jeggeleben (Germany), and Piacenza (Italy). Climate data for 2018–2023 were obtained from the ERA5 reanalysis dataset and included hourly records of global horizontal irradiance (GHI), temperature, relative humidity, wind speed, and precipitation. All irradiance values were cleaned, corrected for low solar angles, and formatted for simulation input ([Setchell, 2020](#)). Further details on data preprocessing and site-specific meteorological variability are provided in Section 1.1 of the Supplementary Methods.

2.2. Simulation framework

We employed the Agri-OptiCE® platform, a coupled simulation and optimisation model developed for APV systems (workflow schematic provided in Supplementary Figures, [Fig. S1](#), Section 4). It integrates (1) a solar irradiance and shading model, (2) a bifacial PV performance model, and (3) a crop growth and microclimate model adapted from EPIC ([Williams et al., 1989](#)), which are coupled with a multi-objective optimisation algorithm. Together, these sub-models simulate the interactions between system layout, local climate, and under-panel microclimate to estimate electricity yield, crop productivity, and water use while simultaneously identifying optimal system designs

under specified constraints. The underlying equations governing radiative transfer, PV conversion, crop growth, and evapotranspiration, as well as the optimisation formulation, are detailed in Supplementary Method (Section 1.2). Model validation against an established tool (*bifacial_radiance*) and experimental data is presented in Supplementary Validation (Section 3).

Shading and PV performance are computed at an hourly temporal resolution, consistent with the ERA5 reanalysis input data, and applied across the six-year crop rotation. The shading model uses view-factor geometry to calculate radiative exchange at the centre of each row, thereby capturing representative under-panel conditions while excluding edge effects. This ensures that results reflect scalable system performance rather than field-boundary artefacts. Simulations were conducted for a six-year crop rotation (ley grass, spring wheat, barley) with location-specific phenology and harvesting dates. Crop model parameters were held constant across sites and derived from literature sources or previous field studies, with full parameter values reported in Supplementary Method (Section 1.2).

2.3. Optimisation approach

A multi-objective genetic algorithm (GA) was used to identify optimal system designs. Key design variables included module orientation, mounting height, row pitch, and system type (vertical, one-axis tracking, and overhead). System performance was evaluated using a composite WEF (water–energy–food) index constructed from normalised objective values. Normalisation was based on Utopia–Nadir scaling to ensure balanced representation of objectives ([Grodzevich and Romanko, 2006](#)). Additional scenarios tested included EF (energy–food), WEEF (water–energy–economy–food), and WEFb, i.e., WEF with enhanced crop model performance (to account for crop adaptation mechanisms using empirically corrected yields). The water term, to be minimised, is based on the ratio of evapotranspiration (ET) under APV to that in open-field conditions. Energy, to be maximised, is represented by the land equivalent ratio for

PV (LER_{pv}), comparing electricity production per hectare under APV to conventional ground-mounted PV. Food, also maximised, is based on LER_{crop} , the ratio of average crop yield over the rotation under APV to that in open-field conditions. Economic performance is maximised and quantified by the net present value (NPV) of the APV system. The full optimisation framework is described in the Supplementary Method: Section 1.3 defines the GA objective functions, Section 1.4 presents the optimisation methodology and design space, and Section 1.5 details the scenario definitions used in this study.

2.4. Performance metrics

APV performance was evaluated using the LER and cumulative ET.

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LER combines relative crop yield (LER_{crop}) and relative PV output (LER_{pv}) per unit land compared to reference cases (open-field crops and conventional PV).

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Water consumption was assessed by comparing crop ET under APV and open-field conditions, serving as a proxy for water use.

All metrics were computed annually and averaged over the crop rotation cycle to account for interannual climate variability. Further description of metric formulation is provided in Supplementary Method, Section 1.6. In addition, a sensitivity analysis of design parameters was performed to visualise how row pitch, height, orientation, and system type influence model outcomes and the optimisation process. This analysis provides a more intuitive understanding of parameter–performance relationships beyond the GA results. Information on the configuration parameters and their ranges is reported in Supplementary Method, Section 1.7.

2.5. Policy constraints

Each scenario is evaluated under three constraint settings derived from real-world regulatory contexts: C1 reflects Swedish subsidy eligibility scheme, imposing a maximum land occupation threshold ([Jordbruksverket, 2024](#)); C2 is based on German regulations and includes both minimum crop yield requirements and land-use limits ([DIN SPEC 91434, 2025a](#)); and C3 represents Italy's more comprehensive framework, integrating provisions from a ministerial decree and sectoral best practices, with requirements on crop yield preservation, land occupation limits, and minimum energy conversion efficiency ([MASE, 2023](#); [MASE, 2022](#)). The constraints applied were as follows:

- C1 (Sweden): Land occupation limit of ≤ 10 %, based on eligibility for agricultural subsidies ([Jordbruksverket, 2024](#)).

- C2 (Germany): Crop yield retention ≥ 66 % and land loss ≤ 15 %, based on DIN SPEC 91434 ([DIN SPEC 91434, 2025b](#)).

- C3 (Italy): Multiple constraints including ≤ 30 % land occupation, ≥ 60 % PV yield compared to conventional systems, elevation ≥ 2.1 m, and ≤ 30 % yield loss (non-binding) ([MASE, 2023](#); [MASE, 2022](#); [PV Magazine, 2022](#); [Svanera et al., 2023](#)).

A detailed breakdown of these constraints and their implementation is provided in Supplementary Method Section 1.8.

3. Results

3.1. Constraints and climate interact to shape optimal system design

Regulatory constraints strongly determine not only the feasibility of solutions but also their geometric characteristics. [Table 1](#) summarises the optimal APV configurations across scenarios, highlighting how parameters such as row pitch, axis height, PV module orientation, and system type shift under each regulatory context. Notably, under the Italian constraints (C3), most objective functions yielded no feasible configuration across the investigated climates, except for WEFb. For C3, feasibility failed primarily because the combined land-use ($\leq 30\%$) and energy thresholds ($\geq 60\%$ of reference PV output) could not be simultaneously satisfied. For instance, at 10 m row pitch the systems delivered $LER_{PV} \approx 0.43\text{--}0.56$, well below the 0.60 threshold. Only in the WEFb scenario, where regression-based yield corrections increased LER_{crop} by approximately 10–30 % compared to the baseline, did the configurations satisfy both agronomic and economic criteria, achieving total LER values up to 1.66 and NPVs approaching €3 million in Italy. This finding underlines how policy strictness can entirely preclude viable APV configurations, even in high-irradiance settings like Italy. A more detailed visualisation of the optimisation process, including Pareto fronts and trade-offs among objectives, is provided in Supplementary Results (Section 2.1).

Table 1. Optimal APV system configurations for each location under different constraint scenarios. Each row shows the optimal combination of row pitch, system height, PV orientation, and configuration type for Sweden, Germany, and Italy under four objective functions (WEF, EF, WEEF, WEFb) and three constraint regimes (C1, C2, C3). Values for LER, LER_{crop} , LER_{PV} , water consumption (WC) represent rotation-averaged performance across the six-year crop sequence.

Scenario	Location (constraint)	Row pitch (m)	Height (m)	PV orientation (°)	System solution	LER_{crop}	LER_{PV}	LER	WC (%)	NPV (k€)
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[illegible]

Scenario

[illegible]

	Italy (C3)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Scenario										
WEEF	Sweden (C1)	10.0	1.5	−75	One-axis	0.69	0.56	1.25	90.4	410
	Germany (C1)	10.0	1.5	−75	One-axis	0.70	0.54	1.24	90.6	792
	Italy (C1)	10.0	1.5	−75	One-axis	0.73	0.54	1.27	86.4	1713
	Sweden (C2)	7.0	0.5	−45	Vertical	0.66	0.65	1.31	90.1	446
	Germany (C2)	8.0	1.5	0	One-axis	0.66	0.63	1.29	89.3	887
	Italy (C2)	7.0	3.0	0	Overhead	0.67	0.69	1.36	83.4	2181
	Sweden (C3)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	Germany (C3)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	Italy (C3)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Scenario										
WEFb	Sweden (C1)	10.0	3.5	−75	One-axis	0.88	0.56	1.44	90.4	216
	Germany (C1)	10.0	3.5	−75	One-axis	0.89	0.54	1.43	90.6	598
	Italy (C1)	10.0	3.5	−90	One-axis	0.92	0.54	1.46	86.4	1518
	Sweden (C2)	7.0	3.5	−60	One-axis	0.75	0.77	1.52	86.6	274
	Germany (C2)	7.0	3.5	−75	One-axis	0.76	0.75	1.51	86.7	808
	Italy (C2)	7.0	3.5	−75	One-axis	0.80	0.75	1.55	80.7	2083
	Sweden (C3)	5.0	3.0	−90	Overhead	0.71	0.84	1.55	84.9	366

Germany (C3)	5.0	3.0	−75	Overhead	0.70	0.88	1.58	84.3	1120
Italy (C3)	5.0	3.0	−30	Overhead	0.70	0.96	1.66	76.8	2998

Moreover, the WEFb scenario demonstrated the highest system performance across all settings, with LER_{PV} reaching up to 0.96 and substantial reductions in water consumption. Relaxing policy restrictions allowed the algorithm to identify system designs that balance shading benefits with energy conversion, increasing both crop yield and NPV. For instance, the optimal configuration in Germany under WEFb-C2 produced a LER of 1.51, with an NPV exceeding €800k.

Interestingly, optimal PV orientation varied not only by latitude but also by constraint type and scenario. Under WEF-C1, the optimisation yielded one-axis tracking systems with slight deviations from east-west orientation: for example, Sweden had -60° , while Germany and Italy used -75° . These subtle shifts aimed to increase irradiance availability for crops by reducing the amount of intercepted light during peak hours, thereby improving under-panel light conditions while modestly sacrificing PV output. Under WEF-C2, which enforced crop yield and land-use thresholds, overhead systems emerged as optimal across all countries. Overhead systems allowed $\sim 10\text{--}20\%$ higher ground-level PAR than tracking layouts at equivalent pitches. This translated into higher LER_{crop} , which was necessary to meet the $\geq 66\%$ crop yield constraint. The modest sacrifice in LER_{PV} was compensated by regulatory compliance. In this configuration, a clear latitudinal trend appeared: Sweden selected -90° (east-facing), Germany -75° , and Italy -15° , progressively aligning closer to south-facing orientations. This reflects a strategic balance between maximising light for crops and maintaining sufficient energy output, especially where irradiance is more abundant.

In contrast, the EF scenario led to more unconventional results. Here, vertical systems became optimal in several cases, most notably when oriented southwards (0°), an unusual choice given the reduction in PV conversion efficiency. However, in the absence of water as a co-optimised resource, the optimiser prioritised increased crop yield, even

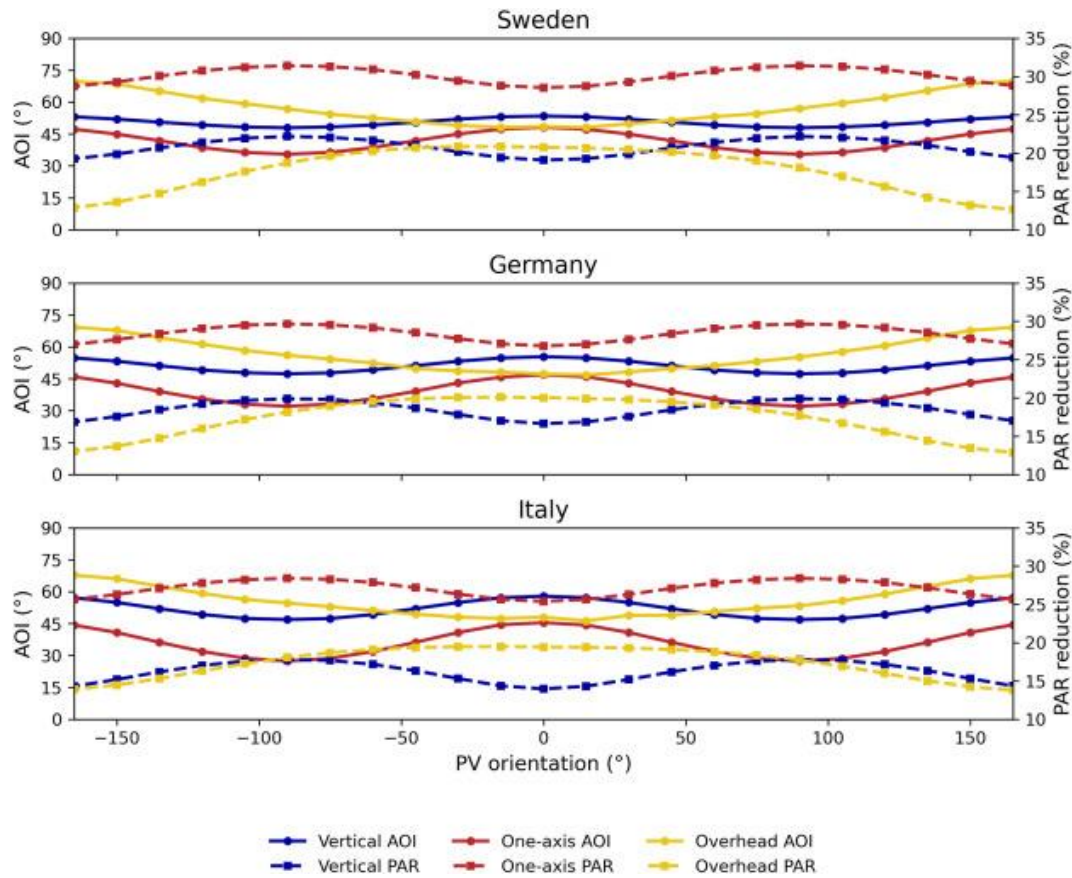
at the expense of PV output. This outcome reflects the trade-off dynamics: while water-saving benefits typically align with greater shading (and thus favour WE synergy), excluding water elevates food production's role in driving system geometry, resulting in more crop-favourable but energy-suboptimal configurations. Two interacting effects explain this trend. First, because energy performance is evaluated using the LER_{PV} (the ratio of APV to reference PV electricity yield per hectare), the optimisation tends to reduce row pitch to install more PV within a given area, even though this increases self-shading and reduces specific yield ($kWh\ kWp^{-1}$). Second, the model may rotate the PV module orientation toward geometries that benefit LER_{crop} more than they penalise LER_{PV} , favouring layouts that improve land-use efficiency but are suboptimal from a techno-economic perspective. These design outcomes highlight the importance of including economic performance in the objective function to ensure that WEF-optimised configurations also meet practical viability thresholds.

Economic considerations strongly influenced optimal configurations. In the WEEF scenario, the inclusion of financial return led to design choices that increased row pitch or shifted systems from overhead to one-axis tracking, thereby raising PV output and revenue, while still meeting agronomic constraints. For instance, in Sweden under C2, the NPV rose significantly from €280k (WEF) to €446k (WEEF), illustrating the impact of revenue-based objectives on spatial layout and system type.

3.2. System trade-offs revealed through correlation analysis

To better understand the factors driving the optimal APV system configurations identified in Section 3.1, we examined the internal trade-offs between LER_{PV} , LER_{crop} , and water consumption using correlation analysis. These results help explain why certain geometries and orientations emerged as optimal, especially under different objectives and regulatory settings. Fig. 1 presents the average angle of incidence (AOI) between incoming solar irradiance and PV modules, along with the corresponding reduction in ground-level PAR for three representative system types: vertical (0.5 m

height), one-axis tracking (2 m height), and overhead (3 m height), each with a fixed row pitch of 10 m. The analysis aggregates crop-season data across all years (2018–2023), reflecting conditions most relevant to plant growth. A clear inverse relationship is observed: configurations with lower AOI exhibit higher PAR reduction, as more sunlight is intercepted by the panels, limiting the light available to crops. In contrast, larger AOI values allow more light to reach the ground, reducing shading. This interaction between PV geometry and irradiance distribution explains why vertical, overhead, and one-axis systems with atypical orientations emerged in our optimisation, PV alignment significantly influences ground irradiance, especially when crop performance is prioritised. Additional analysis in the Supplementary Results (Section 2.2) further shows that the ground coverage ratio (GCR) is strongly correlated with PAR reduction and could be used as a first-order proxy for estimating under-panel light availability. However, this proxy approach has limitations, as it tends to underestimate shading effects for overhead and one-axis systems, indicating that explicit shading simulations remain essential for detailed optimisation.

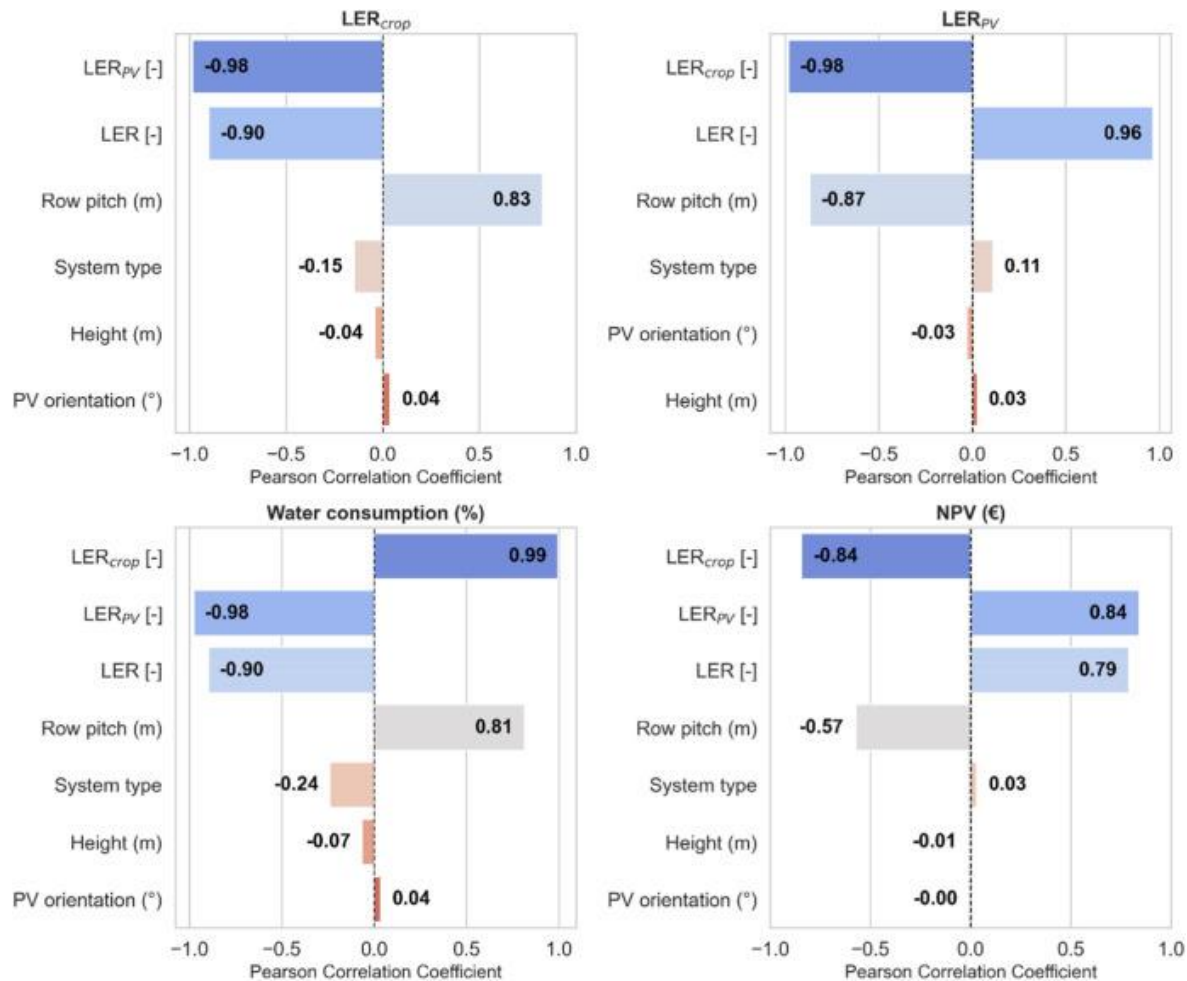


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Fig. 1. Average AOI (solid lines) and corresponding reduction in ground-level PAR (dashed lines) across PV orientations for three APV system types: vertical (0.5 m height), one-axis tracking (2 m height), and overhead (3 m height), all with a 10 m row pitch. Values represent seasonal averages during crop growth periods from 2018 to 2023. Lower AOI values correspond to higher shading and reduced ground-level PAR. To quantify trade-offs across key performance indicators (KPIs), we conducted a correlation analysis of the GA-optimised solutions under WEEF-C1. Fig. 2 shows Pearson Correlation Coefficients between system design parameters and performance metrics including LER, LER_{crop}, LER_{PV}, NPV, and WC. The most dominant trade-off emerged between crop yield and PV output, with a near-perfect negative correlation of -0.98 . LER_{PV} showed a strong positive correlation with overall LER (0.96), whereas

LER_{crop} was negatively correlated (-0.90), suggesting that unconstrained optimisation of LER tends to favour PV electricity conversion per hectare over food production.



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Fig. 2. WEFF-C1 key parameter correlations for Sweden.

Water consumption was also closely linked to system geometry. Row pitch exhibited a strong positive correlation with water consumption (0.81), indicating that more spaced systems reduce shading and thus increase ET. LER_{PV} was negatively correlated with water consumption (-0.98), while LER_{crop} was positively correlated (0.99), again highlighting the central role of shading: designs optimised for specific energy conversion per area reduce water consumption but risk compromising food output. Economic performance (NPV) was found to be strongly influenced by the relative balance between

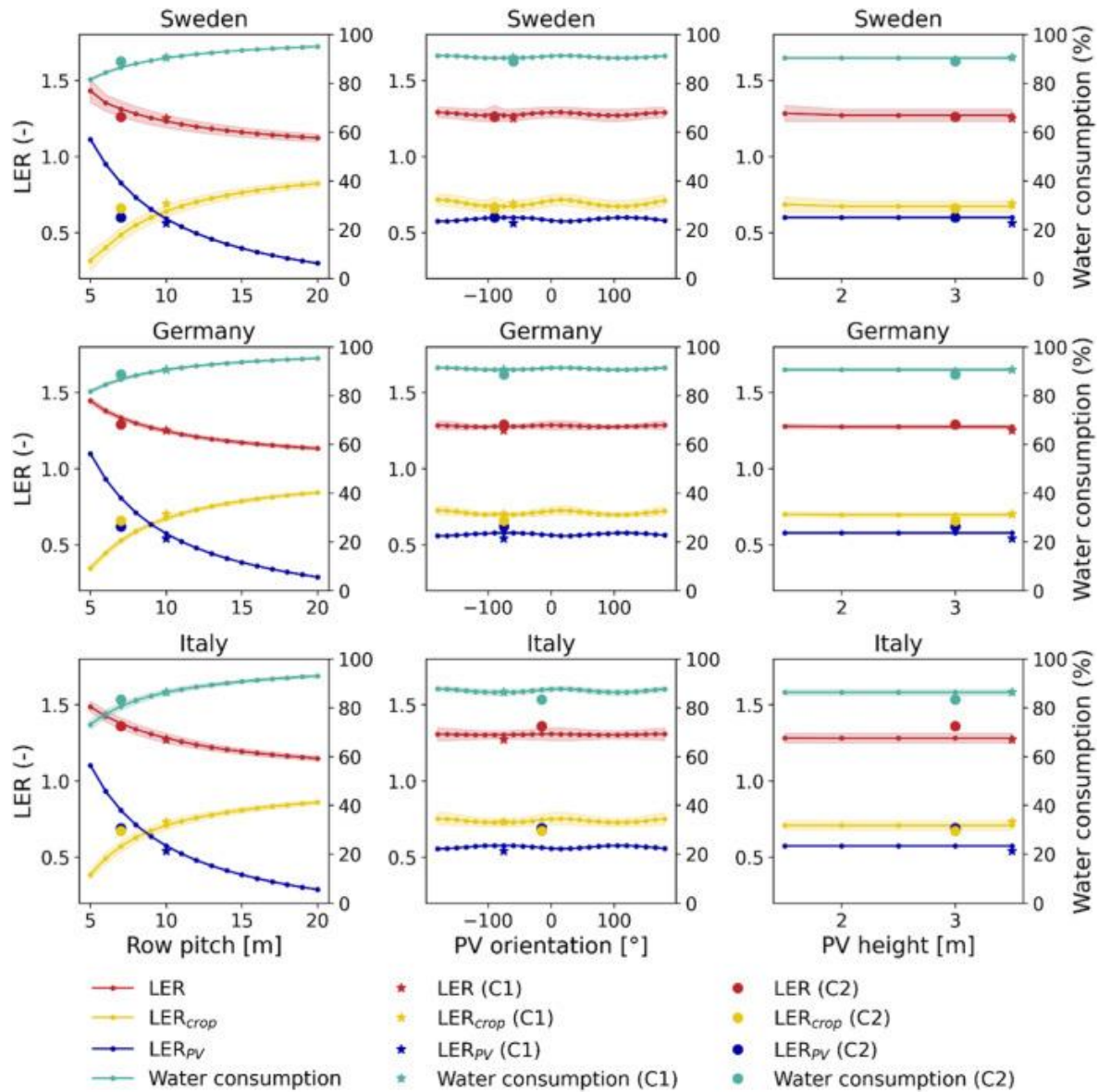
crop and energy yields. NPV was positively correlated with LER_{PV} (0.84) and overall LER (0.79), while negatively correlated with LER_{crop} (−0.84). This indicates that profitability is primarily driven by electricity generation, reflecting the higher economic value of PV compared to crop yields under current market conditions as shown in previous papers (Zidane et al., 2025; Trommsdorff et al., 2023). Among design parameters, row pitch showed a moderate negative correlation with NPV (−0.57), suggesting that denser layouts increase revenue by installing more PV capacity per hectare, even though this reduces crop output and increases shading intensity, and also enhances module self-shading, which in turn negatively affects the specific energy yield. System type, height, and PV orientation played negligible roles in driving NPV in this scenario. These results underscore that economic viability is highly sensitive to the relative weighting of energy versus food production, and that policies aiming to incentivise APV should carefully consider how financial returns are shaped by design geometry.

These trade-offs help explain several outcomes observed in Section 3.1. For example, under the WEF scenario, vertical systems rarely emerged as optimal, as the reductions in LER_{PV} from limited solar capture were often too large to be offset by gains in crop yield, resulting in configurations that underperformed in both energy output and overall WEF performance. In contrast, overhead systems with southward PV module orientations, particularly in Italy, allowed for higher crop performance while still maintaining solid energy conversion, offering a balanced trade-off between food and energy outputs. Furthermore, correlation results underscore the importance of constraint-based design: parameters like row pitch, height, and PV module orientation interact differently across locations and objective functions. For example, in WEF-C1 Sweden, row pitch had a negative correlation with LER (−0.87), consistent with the need to limit land occupation, but this came at the cost of lower yields, a dynamic mirrored in a previous APV study (Campana et al., 2021). A broader comparison of parameter correlations across countries and constraints is provided in the Supplementary Results (Section 2.3), which illustrates how the relative importance of

parameters such as row pitch, orientation, and NPV varies with latitude and policy design.

3.3. Sensitivity of key parameters and interannual variability

To complement the correlation analysis, we performed a univariate sensitivity analysis to explore how key system parameters affect APV performance under the WEF objective. This analysis focused on a one-axis system configuration (east/west-tracking, 10 m row pitch, 2 m height), selected as a reference design. [Fig. 3](#) shows the effects of row pitch, PV orientation, and mounting height on performance indicators, including LER, LER_{crop} , LER_{pv} , and water consumption. Optimal points identified through multi-objective optimisation under C1 and C2 constraints are also indicated.



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Fig. 3. Sensitivity analysis of row pitch, PV orientation, and height on system KPIs.

Results are based on a one-axis east/west-tracking system (10 m pitch, 2 m height).

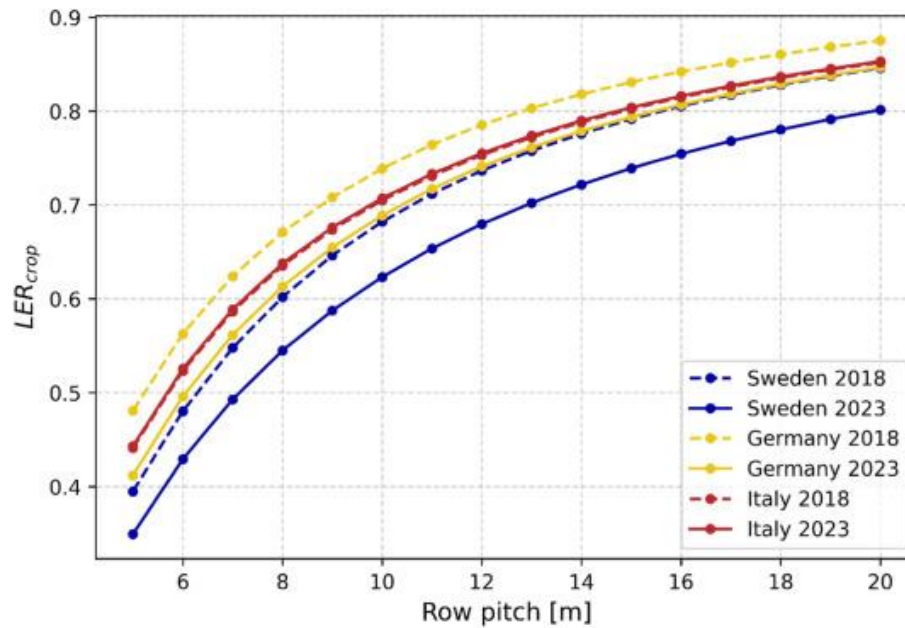
Results correspond to the WEF scenario. Stars indicate optimal solutions identified by the GA under the C1 constraint, while circles represent optimal solutions found under the C2 constraint.

Row pitch was the most influential parameter across all KPIs. Tighter spacing increased LER_{PV} and total LER due to higher PV capacity per hectare, but this came at the expense

of LER_{crop} , reflecting increased shading and reduced light availability for crops. Wider row pitches increased under-panel irradiance and crop productivity, though with diminishing energy returns (vertical and overhead results can be found in Supplementary Results, Section 2.1). These trade-offs are further illustrated by the shaded areas in [Fig. 3](#), which represent interannual variability: systems with narrow row pitch showed larger year-to-year fluctuations in LER_{crop} , particularly in northern climates, underscoring their heightened sensitivity to weather conditions.

PV orientation also had a measurable influence. PV module orientations near 0° (north/south-tracking) produced slightly improved LER_{crop} values. This reflects the role of orientation in shaping daily light distribution and shading dynamics. System height, though less impactful than row pitch, positively influenced light penetration and water use in some cases. These comparisons highlight how system configurations interact with parameter tuning, reinforcing the finding that design trade-offs are configuration-dependent. Further details on the vertical and overhead APV systems, including country-specific LER contributions for each configuration, are provided in the Supplementary Results (Section 2.4). Detailed spatial irradiance distributions generated by the model, including transects across system configurations are provided in previous work ([Zainali et al., 2023](#)).

To assess the effect of year-to-year climate variability on crop performance, we simulated LER_{crop} for spring wheat using a one-axis tracking system (2 m axis height, east–west tracking) in two contrasting weather years: 2018 and 2023, characterised by differences in both solar irradiance and cumulative annual precipitation. As shown in [Fig. 4](#), Italy exhibited stable yields across years, while Sweden and Germany showed substantial variability, particularly at smaller row pitches.



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Fig. 4. Simulated LER_{crop} of a one-axis east–west tracking APV system (2m axis height) with spring wheat in 2018 and 2023 for three countries: Sweden, Germany, and Italy.

This demonstrates that interannual climate effects can rival or even exceed the impact of geometry. Although crop rotation was held constant across years in this study, future work could explore dynamic crop-year allocation, which may reveal even greater performance variability and inform more resilient design strategies. Overall, these results confirm that while spatial configuration determines the structural potential of APV systems, environmental conditions and regulatory thresholds ultimately shape viable design envelopes. Robust optimisation must therefore account for both parameter sensitivities and climatic uncertainty.

4. Discussion

4.1. Trade-offs in system design and performance

Our results reveal that APV system optimisation is shaped by a fundamental trade-off between crop yield and electricity production as also observed in other APV studies ([Trommsdorff et al., 2021b](#); [Bellone et al., 2024](#); [Campana et al., 2021](#); [Miljkovic et al., 2024](#)) and consistent with broader sustainability system integration principles ([Baleta et al., 2019](#)). Across all WEEF-related scenarios, optimising LER consistently favoured electricity production per area (LER_{pv}), as indicated by a near-perfect inverse correlation ($r \approx -0.99$) with LER_{crop} . This indicates that small changes in system layout, particularly row pitch and panel orientation, can shift the balance dramatically between energy yield and food productivity.

However, the strength of this trade-off is also influenced by how LER_{crop} is defined, an issue that remains unresolved in the literature. In this study, we followed the prevailing formulation, where a reduction factor is applied directly to the APV yield to account for unharvestable land near PV rows, while irradiance is averaged across the full row pitch, including both cultivable and obstructed zones. This approach is commonly used ([Willockx et al., 2024](#); [Campana et al., 2021](#); [Mazzeo et al., 2025](#); [Reher et al., 2024](#); [Edouard et al., 2023](#)), but a key limitation is its non-spatial nature. By applying a general percentage penalty to the total yield, it fails to capture the actual land loss, which is concentrated in specific zones occupied by APV infrastructure and its operational margins. This simplification introduces asymmetries, as the opportunity cost of the excluded land, dependent on its specific productivity potential (e.g., varying levels of shading), is not accurately represented. One study has proposed an alternative formulation, where land-use efficiency is instead defined as LER minus a separate land loss term (LL), calculated based on the productivity of the excluded area under open-field conditions ([Trommsdorff et al., 2021b](#)). This more conservative approach offers a clearer estimate of the opportunity cost associated with non-cultivable zones and may be better aligned with

policy-relevant metrics in tightly regulated countries. However, this raises an additional question: should irradiance, and thus productivity, be averaged across the entire field or only over the harvestable area, excluding zones obstructed by infrastructure? The latter would provide a more precise representation of land-use efficiency but requires spatially resolved models capable of capturing irradiance variation across individual grid cells. This increased precision comes at the cost of greater computational complexity. Although we did not implement this formulation in our optimisation, our results underscore the importance of this distinction, and we encourage future studies to carefully assess how LER_{crop} is defined in modelling, especially when comparing results across studies or informing regulatory thresholds. Furthermore, we identified a strong linear relationship between the AOI and the seasonal reduction of ground-level PAR. Lower average AOI values, typically resulting from systems that intercept more direct sunlight, boosted PV yield but also intensified shading. This highlights a key geometric trade-off in APV design, where panel orientation that favours solar capture reduces PAR beneath the array.

The scenario comparisons further demonstrate that including economic metrics (WEEF) shifts the optimisation landscape. In Sweden, the NPV rose from €280k (WEF) to €446k (WEEF) under the same regulatory setting, driven by increased row pitch and a shift from overhead to one-axis tracking systems. These designs enhance revenue from electricity sales while still meeting agronomic thresholds, underscoring the role of economic assumptions in spatial configuration. Yet, our economic inputs were held constant at a single representative value for each country, meaning we did not analyse intra-country variability or scenario-based sensitivity. While CAPEX, OPEX (1 % annually), and electricity price differed between countries ([Zidane et al., 2025](#)), they remained fixed within each national context. Nonetheless, economic outcomes varied across systems and crops due to differences in performance, even though the input parameters themselves did not change. Furthermore, all simulations were conducted for a fixed 1 ha farm layout to ensure comparability across sites and scenarios. As a result, potential effects related to system scale, such as cost reductions, investment thresholds,

and grid connection capacity, were not captured in this analysis. Altogether, system design is deeply sensitive to how objectives are defined and quantified. Modifying weightings (e.g., removing water in the EF scenario) or incorporating crop responsiveness (as in WEFb) produced marked shifts in optimal configurations, illustrating the dynamic and context-dependent nature of APV system performance. Across all scenarios, row pitch emerged as the single most influential design parameter. It governed the trade-off between energy output and under-panel light availability, thereby affecting both land-use efficiency and system feasibility. Optimal spacing typically ranged between 5 and 10 m, consistent with prior APV literature and our own sensitivity results, highlighting its pivotal role in design optimisation.

4.2. Policy constraints and design feasibility

The feasibility of APV designs is highly contingent on the regulatory landscape. In this study, three constraint scenarios were applied: C1 (based on Swedish subsidy eligibility including strict land-use constraint), C2 (derived from German standards including land-use and crop yield constraints), and C3 (reflecting Italian regulations plus best practices). C3 produced no feasible configurations under most objectives, except WEFb, highlighting how overlapping or overly stringent constraints can preclude viable solutions, even in irradiance-rich regions like Italy.

This infeasibility may partly stem from modelling simplifications. The used crop model (EPIC) assumes the maximum leaf area index (LAI) and radiation use efficiency (RUE) as constants, without accounting for physiological crop adaptation to shading. Yet, experimental studies suggest that certain crops can dynamically alter canopy structure, leaf angle, or biomass allocation in response to reduced light availability, thereby mitigating yield penalties ([Marrou et al., 2013](#); [Campana et al., 2024](#)). Because our model does not simulate such plasticity, the resulting yields, especially under partial shading, may be interpreted as conservative worst-case estimates. To address this limitation, an empirically derived regression-based corrections were applied to the baseline EPIC

yields, which adjust for observed shading effects on crop productivity. Incorporating this correction made several C3 configurations feasible, demonstrating that shaded crop calibration is essential for realistic scenario evaluation.

In addition to crop physiology, the choice of crop rotation may also determine whether strict frameworks such as C3 are feasible. Meta-analytical evidence shows that crop species differ markedly in shade tolerance: maize and grain legumes often display strong yield penalties even at modest shading levels, while forages, leafy vegetables, and cereals tend to sustain yields more effectively under reduced PAR ([Laub et al., 2022](#)). Similarly, wheat studies demonstrate that cultivars with greater phenotypic plasticity in leaf area and chlorophyll content can partially compensate for shading stress ([Li et al., 2024](#)). Legumes have also been shown to adjust biomass partitioning and nitrogen fixation under light-limited conditions, buffering yield losses in some contexts ([Umesh et al., 2023](#)). These findings suggest that broadening the crop rotation to include less light-sensitive crops could improve feasibility under stringent constraints like C3, whereas more light-demanding crops would exacerbate infeasibility. Thus, the feasibility of APV systems is not only shaped by system geometry and policy design, but also by agronomic choices that farmers may adapt in practice. A limitation of this study is that such adaptive crop-rotation strategies were not treated as an additional optimisation variable, which could further influence system feasibility and resilience.

Interestingly, differences in system outcomes across countries were often more strongly influenced by the type of constraint than its nominal strictness. For example, although C2 explicitly enforced a crop yield constraint, C1 (which did not) resulted in higher LER_{crop} . This outcome reflects C1's more restrictive land occupation limit, which favoured wider row spacing and improved crop light conditions, albeit at the cost of lower LER_{PV} and reduced overall land use efficiency. This highlights how spatial constraints can indirectly enhance crop outcomes while compromising energy yield, underscoring the importance of carefully balancing policy targets.

A further assumption with policy relevance involves the treatment of productivity-linked land value (PLV) (MASE, 2022). We assumed a 1:1 relationship between crop yield reduction and land value loss, as a proxy for incentive scheme eligibility (e.g., under Italy's PNRR (MASE, 2022)). While this approach provides a tractable method for integrating policy effects into optimisation, it does not capture the full market or branding value of crops produced under APV, particularly for premium or sustainability-labelled products like wine grapes (Escibano et al., 2020). Moreover, constraint C3 combined formal regulatory requirements with best practice recommendations, resulting in a particularly narrow design space. The inclusion of both a crop yield constraint and a land occupation threshold illustrate how overlapping criteria, each reasonable in isolation, can jointly preclude viable solutions. Interestingly, when the crop yield constraint is removed from C3 while retaining the land coverage and energy thresholds, the scenario becomes less restrictive than either C1 or C2. This underscores the importance of carefully calibrating constraint design to reflect realistic trade-offs. Overly compound or rigid criteria may fail to accommodate regional differences, technological adaptation, or practical implementation pathways. A more effective policy approach may lie in prioritising key constraints (e.g., land occupation) while allowing flexibility in others, thereby preserving the intent of agricultural protection without unduly limiting system innovation.

4.3. Climate robustness and generalisation

Despite incorporating multi-year weather data and a consistent six-year crop rotation, our results showed relatively limited interannual variability in KPIs across most scenarios. This apparent robustness was most pronounced in Italy, where LER_{crop} for spring wheat remained stable between 2018 and 2023, even as Sweden and Germany exhibited more pronounced year-to-year fluctuations. These differences appear linked to interannual precipitation variation and ET rates, with more humid and variable climates showing greater sensitivity to weather conditions. However, this perceived stability is also a function of model structure and simplifying assumptions.

In addition, all fixed-tilt systems (except vertical) were simulated using a standard 35° tilt angle, a value typical for European PV installations. Although this offers a broadly suitable average, it may not reflect optimal performance at specific latitudes ([Ferry et al., 2025](#); [Saint-Drenan et al., 2018](#)). Similarly, a uniform albedo value of 0.2 was applied for all crop surfaces, despite known differences between species that can affect bifacial PV output ([Sieber et al., 2022](#)). Moreover, a constant ratio of $\text{PAR}/\text{GHI} = 0.45$ was assumed across all locations, even though atmospheric characteristics likely vary between Sweden, Germany, and Italy, leading to site-dependent differences in this ratio. These generalisations reduce spatial specificity but allow broader insight into system-level trends across Europe.

Our sensitivity analyses show that system geometry plays a key role in mediating interannual robustness. Narrower row spacing and south-facing PV orientations increased shading, which in turn led to greater year-to-year variation in LER and LER_{crop} , particularly in Sweden and Germany. These configurations created stronger irradiance gradients under the panels, making system performance more sensitive to annual weather fluctuations. In contrast, systems with wider row pitches or more uniform light distribution (e.g., east–west one-axis tracking) tended to produce more stable outcomes across years, albeit sometimes at the expense of overall energy yield. Altogether, these results highlight the importance of considering both biophysical design and model realism in evaluating APV system robustness. While the crops used here performed stably under tested conditions, this outcome may not generalise to shade-sensitive species or to sites with more extreme climate variability. Future research should expand the crop set, integrate dynamic models capable of simulating phenotypic adaptation, and explore climate stressors beyond the historical record to fully assess long-term performance and resilience under climate change.

5. Conclusions

This study presents a multi-objective optimisation framework for evaluating APV systems across three European locations with varying climate conditions and regulatory constraints. The optimisation integrates water, energy, economy, and food objectives using a normalised weighted sum method, which converts multiple objectives into a single linear performance index. This allows for systematic exploration of trade-offs and priorities under different stakeholder perspectives and policy scenarios. The key findings are:

- Row pitch was the dominant design variable, with optimal spacing consistently between 5 and 10 m across locations and scenarios. Under the baseline crop model, pitches <7 m frequently violated yield/coverage constraints, whereas under WEFb 5 m overhead solutions became feasible. Pitches >10 m reduced land-use efficiency and energy returns per hectare.

- A strong trade-off between PV output and crop yield was confirmed in Sweden-C1 ($r = -0.98$). Water consumption correlated positively with crop yield (0.99) and negatively with PV output (-0.98), indicating that energy-optimised designs reduce ET via shading but at the expense of crop productivity.

- A strong linear relationship was found between annual mean AOI and ground PAR reduction. Orientations that intercept more direct irradiance increased PV yield but reduced under-panel PAR, illustrating a core geometric trade-off.

- Differences between scenarios were shaped more by the type of constraint than by its strictness. Under C1, one-axis with wider spacing (~10 m) best balanced electricity

conversion and eligibility. Under C2, overhead systems with 7–8 m spacing were the only consistently feasible solutions due to the $\geq 66\%$ yield-retention requirement.

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The combined Italian guidelines and best practices constraints (C3) applied several strict constraints, including minimum PV output, maximum land occupation, allowable yield loss, and a system height limit. This led to no feasible solutions across sites.

However, when applying the WEFb scenario, where shading effects were corrected using regression models derived from a recent meta-analysis (relating crop yield to shading rate), feasible solutions emerged. This demonstrates that scenario feasibility is highly sensitive to how crop responses to shading are represented. Rigid regulatory thresholds may therefore risk over-constraining system design unless supported by crop models that explicitly account for shading responses, ideally through empirically validated and integrated APV modelling platforms.

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Interannual variability in KPIs was generally low, aided by a balanced rotation including shade-tolerant crops and by configurations that tempered year-to-year irradiance swings. Variability increased with narrower pitches and less conventional orientations. This underscores the need for improved crop-parameter calibration under APV, ideally with multi-year, multi-environment datasets capturing shaded and open-field dynamics.

Overall, these results provide quantitative design guidance under specific assumptions on feasible APV configurations and show that policy design critically shapes the viable envelope: overly rigid compound constraints can eliminate solutions, while well-targeted, evidence-based rules leave room for context-sensitive optimisation.

CRedit authorship contribution statement

Sebastian Zainali: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation,

Conceptualization. **Silvia Ma Lu:** Writing – review & editing, Validation, Methodology. **Yuri Bellone:** Writing – review & editing, Writing – original draft, Methodology, Data curation. **Pietro Elia Campana:** Writing – review & editing, Writing – original draft, Validation, Supervision, Methodology, Funding acquisition, Data curation, Conceptualization.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: The company European Energy is financing half of Sebastian Zainali's Ph.D. salary.

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Appendix A. Supplementary data

The following is the Supplementary data to this article: [Download: Download Word document \(6MB\)](#)

Multimedia component 1.

Data availability

Data will be made available on request.